

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 03)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate

☐

B The readings are accurate but not precise

☐

C The readings are both accurate and precise

☐

D The readings are reliable because the meter is electronic

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0 1 . 2

Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

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0 1 . 3

Explain why a preliminary study is carried out before the main investigation.

[2 marks]

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0 2 . 1

State the Brundtland Commission definition of sustainable development.

[1 mark]

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0 2 . 2

Explain the difference between a negative feedback mechanism and a positive feedback mechanism, giving **one** example of each in the climate system.

[4 marks]

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0 2 . 3

Explain why a diverse ecosystem is more likely to resist change than an agroecosystem.

[1 mark]

0 2 . 4

Evaluate the extent to which copying the principles of natural systems could make human activities more sustainable.

[9 marks]

Do not write outside the box

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15

03 . 1

Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

- A Thermal pollution of a river by a power station
- B Flooding a valley to create a reservoir
- C Lowering the water table by groundwater abstraction
- D Acid mine drainage entering a stream

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03 . 2

Give **four** human activities that cause complete habitat destruction.

[4 marks]

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04 . 1

Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

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Explain why growing coral reefs help to resist climate change.

[2 marks]

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Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

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1

 Complete Table 1.

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Method	Target species	One environmental disadvantage	
Demersal trawling			
Purse seining			
Pelagic long-lining			

[2 marks]

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0 5 . 3

Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

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0 6 . 1

Give **two** natural processes that are driven by solar energy.

[2 marks]

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0 6 . 2

Explain why natural processes can take place at low temperatures.

[2 marks]

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0 6 . 3

Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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07 . 1

Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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07 . 2

Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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07 . 3

Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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08

A new pond was dug in a nature reserve. Table 1 shows the vegetation recorded at intervals over 60 years.

Table 1

Years after creation	Dominant vegetation	Water depth at centre / m
1	Single-celled algae, floating duckweed	2.0
5	Submerged pondweeds; reeds at margins	1.8
15	Reeds and reedmace over most of pond; water lilies	1.0
30	Marsh plants, sedges; willow and alder at edges	0.2
60	Willow / alder carr; oak seedlings	0 (soil waterlogged)

08 . 1

Name the type of sere shown in Table 1 and identify the pioneer species.

[2 marks]

08 . 2

Explain the changes shown in Table 1.

[5 marks]

08 . 3

Explain why a lithosere and a hydrosere in the same region end with the same climax community.

[2 marks]

08 . 4

Describe the early stages of a lithosere.

[1 mark]

10

0 9 Table 1 shows some designated protected areas in the UK.

0 9 . **1** Complete Table 1 to show the legislation under which each area is designated and its key feature.

[4 marks]

Table 1

Designation	Designated under	Key feature
Site of Special Scientific Interest (SSSI)		
Special Protection Area (SPA)		
Ramsar site		
Marine Conservation Zone (MCZ)		

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0 9 . **2** Explain how designating a protected area protects habitats and species.

[3 marks]

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0 9 . **3** Explain why voluntary agri-environmental schemes are needed in addition to legislation.

[3 marks]

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1 0 Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

Year	Adults caught in traps	Ponds with newts detected by eDNA (out of 20)
2019	14	6
2020	22	9
2021	18	11
2022	31	13
2023	27	15

1 0 . **1** Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

Answer %

1 0 . **2** Explain how eDNA detects the presence of newts without catching them.

[2 marks]

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10 . 3

Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

10 . 4

Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

[9 marks]

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Write an essay on one of the following topics.

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Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

OR

1 1 . 2

Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Handwriting practice area with 25 horizontal dotted lines.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

END OF QUESTIONS